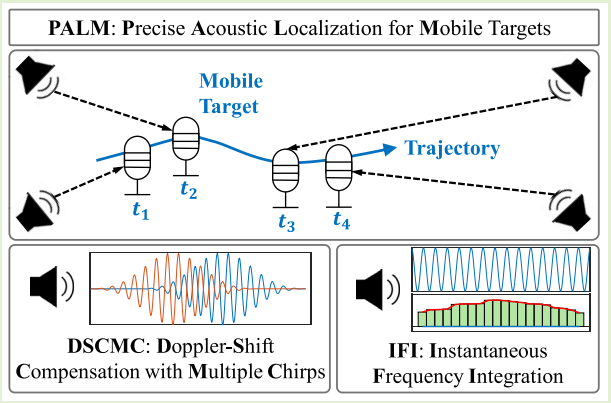
PALM: Precise Acoustic Localization for Mobile Targets With Doppler-Shift Compensation and Relative Distance Estimation

Yuzhang Xi, Weimeng Cui, Guangyao Liu, and Zhi Wang, *Member, IEEE*

***Abstract*—Acoustic indoor positioning systems (AIPSs) offer advantages such as low cost and high accuracy. However, when tracking moving targets, their accuracy significantly degrades due to the Doppler effect and long signal reception intervals (SRI). To address the above issue, we present the precise acoustic localization for mobile (PALM) targets solution. PALM first obtains an unbiased time of arrival (ToA) estimation using the proposed Doppler-shift compensation with multiple chirps (DSC-MC) method. It then compensates for the relative displacement between signals using the proposed instantaneous frequency integration (IFI) algorithm. Finally, PALM estimates the target's position based on time difference of arrival (TDoA) and maximum likelihood (ML) estimation. Simulations demonstrate the superior performance of the proposed IFI and DSC-MC methods and present their performance under different parameter settings. Extensive experiments demonstrate the real-world accuracy of the proposed algorithms and verify that the PALM system achieves centimeter-level positioning accuracy and optimal algorithmic performance in mobile target localization scenarios.**



***Index Terms*— Acoustic indoor positioning, chirp signal, Doppler-shift compensation (DSC), instantaneous frequency (IF), time difference of arrival (TDoA).**

I. Introduction

INDOOR localization technologies provide essential information for various targets to enable perception and interaction with the external environment. Different sensor networks, such as ultrawideband (UWB) [1], Bluetooth [2], Wi-Fi [3], and acoustic systems [4], are suited to different application scenarios due to variations in localization algorithms, signal schemes, and hardware configurations [5].

Among these technologies, acoustic indoor positioning systems (AIPSs) have attracted widespread research interest due to their advantages of low cost and high accuracy [6]. However, most AIPS applications target low-speed objects such as pedestrians and smartphones. When it comes to high-speed targets with stringent localization accuracy requirements, such as mobile robots, autonomous vehicles, and drones, acoustic systems are rarely adopted, whereas UWB systems, which offer comparable accuracy, are widely used [7]. This disparity largely stems from the inherently low propagation speed of sound.

On one hand, the slow speed of sound enables precision distance estimation [8]. Compared with electromagnetic waves, sound waves can provide more accurate range measurements under similar time of arrival (ToA) or time difference of arrival (TDoA) estimation accuracy. Numerous studies have demonstrated that acoustic-based distance estimation can consistently achieve centimeter-level accuracy, even in challenging environments [9], [10], [11], [12]. On the other hand, the low propagation speed significantly degrades the localization accuracy for moving targets in acoustic systems, due to both direct and indirect impact.

First, the direct impact arises from the well-known Doppler effect. Specifically, the Doppler frequency shift leads to biased estimates in ToA and TDoA. Recently, many studies have focused on mitigating these Doppler-induced errors through Doppler-shift compensation (DSC) algorithms [13], [14], [15], [16]. However, most of these algorithms only achieve approximate compensation in theory and fail to completely eliminate

Received 20 July 2025; accepted 11 September 2025. Date of publication 23 September 2025; date of current version 31 October 2025. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB3900800. The associate editor coordinating the review of this article and approving it for publication was Prof. Jun Wang. *(Corresponding author: Zhi Wang.)*

The authors are with the State Key Laboratory of Industrial Control Technology, Zhejiang University, Hangzhou 310027, China, and also with Huzhou Institute, Zhejiang University, Huzhou 313000, China (e-mail: zjuwangzhi@zju.edu.cn).

Digital Object Identifier 10.1109/JSEN.2025.3611265

1558-1748 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

the influence of velocity, thus preventing truly unbiased estimation. Furthermore, to achieve DSC, the existing algorithms require extending the time–frequency occupancy (TFO) of signals, which reduces the system flexibility and exacerbates the errors introduced in the next issue.

Second, the indirect impact causes the mobile target to be at different positions when receiving signals from different anchors. The varying propagation delays of signals from different anchors create large signal reception intervals (SRI), during which the target's position has already changed. In addition, due to the limited bandwidth in AIPSs, time-division multiple access (TDMA) is often used to prevent signal interference, introducing additional delays known as anchor guard intervals (AGI) [17]. And greater TFO leads to greater AGI, which further increases SRI and localization errors. However, existing studies have overlooked this problem because it is negligible for low-speed targets. For example, with an SRI of 0.2 s, a target moving at 0.2 m/s only displaces by 0.04 m; however, when the target speed reaches 2 m/s, the displacement becomes 0.4 m, which is unacceptable for systems aiming for centimeter-level accuracy.

To address the aforementioned challenges and enable centimeter-level acoustic localization for moving targets, we propose the precise acoustic localization for mobile (PALM) solution: PALM target. The key innovations and contributions of this work are summarized as follows.

1) The PALM solution first using the proposed DSC method achieves high-precision ToA estimation. Then sinusoidal signals are introduced to compensate for target displacement during the SRI. Finally, accurate localization of moving targets is achieved through the modified TDoA-ML algorithm.
2) To obtain the relative displacement estimation (RDE) of the target during SRI, we propose the instantaneous frequency integration (IFI) algorithm. The IFI algorithm derives the instantaneous frequency (IF) of sinusoidal signals using the Hilbert transform (HT), and then integrates the difference between the instantaneous and fixed frequencies to estimate relative displacement based on the Doppler effect. This method achieves extremely high RDE accuracy under strong noise and varying time intervals.
3) We propose a novel Doppler-shift compensation with multiple chirps (DSC-MC) method, which enables theoretically unbiased DSC with arbitrary distinct chirp signals. The signal schemes proposed in this article can achieve minimal or no increase in TFO.
4) Extensive simulations and real-world experiments demonstrate: first, the IFI algorithm achieves state-of-the-art performance in RDE and second, various signal designs with DSC-MC method yield superior DSC results. Experiments also show that PALM achieves an average localization accuracy of 0.0582 m for moving target positioning at an average speed of 1 m/s, exhibiting superior performance.

The remainder of this article is organized as follows. Section II introduces related work. Section III provides a detailed description of the PALM positioning method, along with the IFI and DSC-MC algorithms it uses. Section IV presents the simulation results of the IFI and DSC-MC methods under different parameter settings. Section V evaluates the performance of each algorithm and the overall PALM system through experiments. Finally, Section VI summarizes this article.

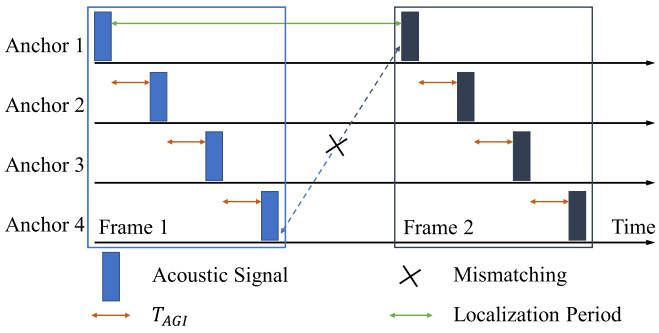


Fig. 1. Signal timing and localization framework of AIPSs.

II. Related Works

A. AIPS Architecture

AIPSs have been extensively studied, and many mature solutions have already found practical applications. These systems typically deploy anchor nodes at the vertices of a rectangular localization area, which sequentially transmit chirp signals at regular intervals. Based on ToA or TDoA measurements, target localization is achieved. The signal timing and localization framework are illustrated in Fig. 1. AIPSs such as [14], [17], [18], [19], [20], [21], [22], and [23] conform to or resemble this architecture. The proposed PALM solution is also built upon this architecture.

B. Relative Displacement Estimation

To compensate for the error caused by the target's movement during the SRI, it is necessary to obtain the target's displacement relative to the anchor over a given time interval. This is referred to as the RDE problem.

In earlier work, Mao et al. [24] used an IMU sensor to first obtain the absolute displacement, which was then projected to derive the RDE. However, since the projection process itself requires position information, this approach is complex and lacks high accuracy.

A more convenient strategy is to estimate displacement directly by signal processing. The simplest method involves using the Fourier transform (FT) or short-time Fourier transform (STFT) to calculate the frequency shift caused by motion, from which the displacement can be inferred. However, when the time interval is short, the frequency resolution of FT becomes very low. Another approach is to measure phase changes. The LLAP algorithm [25] decomposes a sinusoidal signal into its I/Q components to track phase variation, with an accuracy upper bound determined by the phase change within the sampling interval. The Vernier algorithm [26], [27] leverages the mismatch between the sampling rate and the signal frequency, following the principle of a vernier caliper,

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

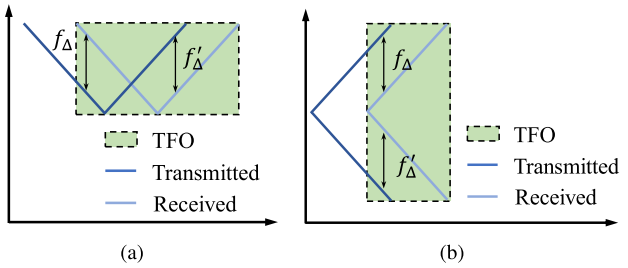


Fig. 2. Signals of conventional postprocessing-based DSC methods. (a) Downchirp extended before an upchirp. (b) Downchirp extended above an upchirp.

to achieve resolution beyond the sampling rate limit, reaching a theoretical precision of 0.17 mm [26].

However, these RDE algorithms are mostly used for short-range gesture recognition or target tracking and have not been applied to AIPSs.

C. Doppler-Shift Compensation

There are two main approaches for performing DSC on the chirp signals transmitted by AIPSs: preprocessing and postprocessing.

The preprocessing approach first estimates the frequency shift caused by the Doppler effect, and then adjusts the reference signal accordingly before estimating the ToA. When the target moves, the time–frequency slope of the linear frequency modulation (LFM) signal changes. This slope can be estimated using the fractional FT (FrFT). In [28], the slope is estimated by iteratively searching for the peak in the FrFT domain. This method is further extended to cover multiple targets and both the near-field and far-field conditions [29]. However, the drawback of such methods is their high complexity, and even a small slope estimation error can lead to significant errors in distance estimation.

The postprocessing approach first obtains the ToA estimation, and then applies compensation. The underlying principle is that chirp signals with positive and negative slopes produce opposite estimation errors, and taking the average of the two can effectively suppress the error. In [13], a downchirp is extended before an upchirp, forming the signal pattern shown in Fig. 2(a). In [14] and [15], the downchirp is extended above the frequency band of the upchirp, as shown in the signal scheme in Fig. 2(b). However, the above method increases the TFO, and the resulting estimation is still an approximation.

III. Methodology

In this section, we will provide a detailed explanation of the localization process of PALM and the working principles of two novel algorithms, IFI and DSC-MC.

A. PALM Overview

First, the signal design and workflow of PALM are illustrated in Fig. 3. Initially, the chirp signals are combined with a sine signal and subsequently emitted by the speaker. Upon receiving the signal, the microphone captures it, after which the system isolates the target signals across different frequency

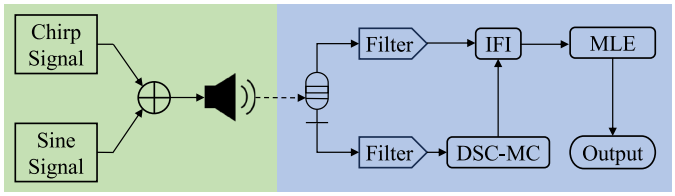


Fig. 3. Workflow of the PALM system.

bands using bandpass filters. Once the DSC-MC algorithm estimates the ToA results of the chirp signals, it determines and provides the appropriate SRI window for the IFI algorithm. The IFI algorithm then compensates for the relative displacement of the target within the specified window and outputs the corrected TDoA results. Based on these TDOA results, the system computes the estimated position of the target.

It is important to note that the passband characteristics of the bandpass filter differ for chirp signals and sinusoidal signals. For a narrowband sinusoidal signal with a frequency f, the corresponding passband is set as $[f - f_{\text{bp}}/2,\ f + f_{\text{bp}}/2]$, where $f_{\text{bp}} = 2v_{\max} f/c$, and $v_{\max}$ represents the maximum target velocity. In this study, $v_{\max}$ is 2 m/s. This passband design aims to preserve signal integrity while enabling effective signal separation. In contrast, for wideband chirp signals, the filter passband should align with the bandwidth occupied by the transmitted signal.

Similar to conventional AIPSs, PALM performs localization using four anchor nodes based on TDoA. The key difference is that PALM introduces a sinusoidal signal to estimate the relative displacement between received signals. Fig. 4 illustrates the localization error caused by ignoring RDE, as well as the displacement that needs to be compensated.

Assume the target receives signals from anchors 1, 2, 3, and 4 sequentially during its motion, with their respective arrival times denoted as t_1, t_2, t_3, t_4. Since the localization algorithm can only be executed after receiving the fourth signal, what the algorithm actually estimates is the target's position at t_4, denoted as $\mathbf{s}_4$. However, due to the continuous movement, the time at which each signal reaches position $\mathbf{s}_4$ is not the same as the observed arrival time. For anchor i ($i = 1, 2, 3$), the actual ToA at position $\mathbf{s}_4$ is

$$t'_i = t_0 + \frac{\|\mathbf{s}_4 - \mathbf{a}_i\|_2}{c} = t_i + \frac{\mathcal{D}_{4i}}{c}. \tag{1}$$

In (1), $\mathbf{a}_i$ represents the position of anchor i, c is the speed of sound, $\|\cdot\|_2$ refers to the ℓ_2-norm, and $\mathcal{D}_{4i}$ denotes the distance difference between positions $\mathbf{s}_4$ and $\mathbf{s}_i$ relative to anchor position $\mathbf{a}$, i.e., the unknown relative displacement.

To estimate the position, at $\mathbf{s}_4$, the actual TDoA is

$$\mathcal{T}_{4i} = \frac{1}{c}\left(\|\mathbf{s}_4 - \mathbf{a}_4\|_2 - \|\mathbf{s}_4 - \mathbf{a}_i\|_2\right). \tag{2}$$

The estimated TDoA is calculated as the difference between the compensated observed ToA values, minus the delay caused by the AGI

$$\hat{\mathcal{T}}_{4i} = \hat{t}_4 - \hat{t}'_i - (4-i)\,T_{AGI} = \hat{t}_4 - \hat{t}_i - \frac{\hat{\mathcal{D}}_{4i}}{c} - (4-i)\,T_{\text{AGI}}. \tag{3}$$

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

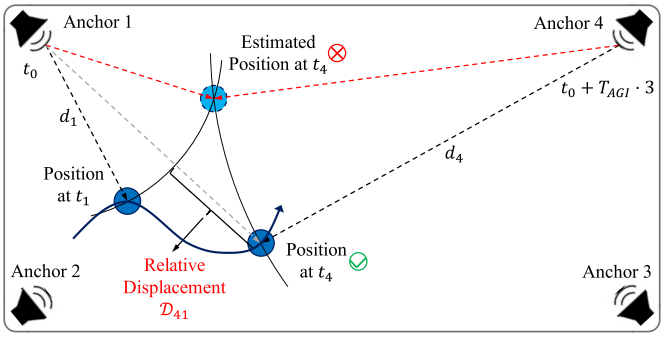


Fig. 4. Relative displacement and position estimation error caused by target movement.

Thus, the maximum likelihood (ML) estimation equation based on TDoA can be expressed as

$$\mathbf{P}\left(\hat{\mathcal{T}}_{4i}|\mathbf{s}_4\right) = \prod_{i=1}^{3} \frac{1}{\sqrt{2\pi}\sigma} \exp - \frac{\left(\hat{\mathcal{T}}_{4i} - \mathcal{T}_{4i}\right)^2}{2\sigma^2}. \tag{4}$$

Then the solution for the position is

$$\begin{aligned} \hat{\mathbf{s}}_4 &= \arg\max_{\mathbf{s}_4} \ln \mathbf{P}\left(\hat{\mathcal{T}}_{4i}|\mathbf{s}_4\right) \\ &= \arg\min_{\mathbf{s}_4} \sum_{i=1}^{3} \left(\hat{\mathcal{T}}_{4i} - \mathcal{T}_{4i}\right)^2. \end{aligned} \tag{5}$$

In both (4) and (5), $\hat{\cdot}$ stands for the corresponding estimation value.

It can be seen that after compensating for the relative displacement, PALM achieves a more reasonable position estimation result. However, there are still two critical estimates that affect the accuracy of the result. First, the RDE $\hat{\mathcal{D}}_{4i}$. Since this value is relatively small, we must obtain extremely accurate estimation result to avoid introducing larger errors during compensation. Second, the ToA $\hat{t}_i$ of the moving target. This value determines the time window of the RDE, and any error in ToA estimation will introduce errors into the RDE. Therefore, an unbiased estimation of ToA t_i is essential.

To achieve these two goals, we propose the IFI algorithm and the DSC-MC method, respectively.

B. RDE Based on IFI

As shown in Fig. 4, the time interval of the signals transmitted by anchor i and 4 to reach the target is

$$\Delta t_{4i} = t_4 - t_i = (4 - i)\, T_{\text{AGI}} + \frac{d_4 - d_i}{c}. \tag{6}$$

During the time interval $[t_i, t_4]$, we assume that the target velocity in the radiation direction of the anchor i is $v(t)$. Hence, the relative displacement can be expressed as

$$\mathcal{D}_{4i} = \int_{t_i}^{t_4} v\,(t)\, dt. \tag{7}$$

It is worth to note that the SRI in (6) is not the TDoA described in (2). And the RDE is influenced not only by the known AGI but also by the unknown positions and velocities of the target.

Unlike traditional methods that estimate velocity using STFT, or directly estimate displacement from phase changes with moving windows, the proposed IFI algorithm first applies the HT to obtain the IF, then estimates the instantaneous velocity from the frequency shift, and finally computes the displacement. The advantages of this method will be demonstrated below.

In the PALM system, anchors periodically transmit chirp signals and simultaneously emit continuous sinusoidal signals. For a sinusoidal signal with frequency f and wavelength $\lambda = c/f$, the signal received by the target during time interval $[t_i, t_4]$ is

$$u\,(t) = \cos 2\pi \frac{c - v\,(t)}{\lambda} t = \cos 2\pi f_I\,(t)\, t, t \in [t_i, t_4]. \tag{8}$$

When the velocity changes over time, $f_I(t)$ represents the IF of $u(t)$. To achieve the estimation of IF, we use Gabor's method [30] by performing the HT to get the complex signal from the real one. The complex signal is

$$z\,(t) = u\,(t) + j\mathrm{H}\left[u\,(t)\right] = a\,(t)\, e^{j\phi(t)} \tag{9}$$

where $z(t)$ is Gabor's complex signal, and $a(t)$ and $\phi(t)$ are the envelope and phase of $z(t)$, respectively. And H[·] is the HT operator

$$\mathrm{H}\left[u\,(t)\right] = p.v. \int_{-\infty}^{+\infty} \frac{u\,(t-\tau)}{\pi\,\tau} d\tau \tag{10}$$

where $p.v.$ denotes the Cauchy principle value of the integral [31]. Then the IF of $u(t)$ can be calculated

$$f_I\,(t) = \frac{1}{2\pi} \frac{d\phi}{dt}. \tag{11}$$

Due to the influence of noise, the actual estimated IF $\hat{f}_I(t)$ contains noise $f_e(t)$, expressed as

$$\hat{f}_I\,(t) = f_I\,(t) + f_e\,(t). \tag{12}$$

Using the Doppler frequency shift formula

$$\Delta f = \frac{v}{c} f \tag{13}$$

the estimated instantaneous velocity at the radiation direction can be obtained

$$\hat{v}\,(t) = \frac{f - \hat{f}_I\,(t)}{f} c. \tag{14}$$

It is worth noting that we define the velocity away from the anchor as positive here, consistent with the sign convention used in (1). And substituting (12) and (14), the RDE then is

$$\hat{\mathcal{D}}_{4i} = \int_{t_i}^{t_4} \hat{v}\,(t)\, dt = c\Delta t_{4i} - c\frac{\int_{t_i}^{t_4} f_I\,(t)\, dt + \int_{t_i}^{t_4} f_e\,(t)\, dt}{f\,\Delta t_{4i}}. \tag{15}$$

Generally, assuming the noise is additive white Gaussian noise, and since this noise has a uniform power spectral density (PSD), its frequency distribution satisfies

$$f_e\,(t) \sim U\,(-f_s/2,\, f_s/2) \tag{16}$$

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

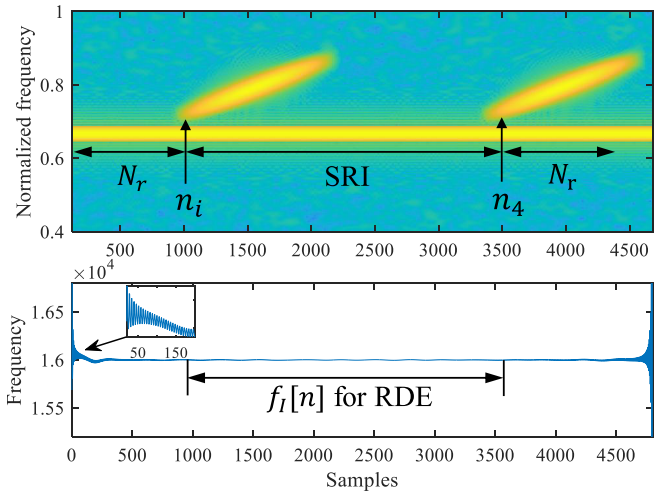


Fig. 5. Spectrum of the received signal and IF of the sinusoidal component.

where U stands for the uniform distribution, and f_s is the sampling frequency. Thusly, when the SRI is long enough, we have

$$\int_{t_i}^{t_4} f_e(t)\,dt = \Delta t_{4i} \cdot E\left[f_e(t)\right] = 0. \tag{17}$$

In this case, the estimation of displacement can be equal to the true value

$$\hat{\mathcal{D}}_{4i} = c\Delta t_{4i} - c\int_{t_i}^{t_4} \frac{f_I(t)}{f} dt = \int_{t_i}^{t_4} v(t)\,dt = \mathcal{D}_{4i}. \tag{18}$$

However, in practice, we are dealing with discrete values, so the IF is obtained as follows:

$$\hat{f}_I[n] = \frac{1}{2\pi}\phi[n] * \delta[n] + f_e[n] \tag{19}$$

where n is the samping index, $\delta(n)$ is the impulse response of the discrete finite impulse response (FIR) differentiating filter, and (*) denotes convolution in time. The detail of this process was discussed in [32]. As a result, in practical applications, the expression for the RDE is

$$\hat{\mathcal{D}}_{4i} = \frac{c}{f_s} \cdot \sum_{n_i}^{n_4} \left(1 - \frac{\hat{f}_I[n]}{f}\right) \tag{20}$$

where n_i represents the discrete value of the estimated ToA t_i, and $n_i = t_i \times f_s$.

In addition, to mitigate the impact of window truncation on IF estimation, redundant length N_r should be added before and after the estimation interval. Assuming signal length N, the HT step has complexity $O(N \log N)$, and subsequent steps are $O(1)$. Thus, IFI's total complexity is only $O(N \log N)$.

In summary, as shown in Fig. 5, the IFI algorithm first performs bandpass filtering on the signal, retaining only the sinusoidal component. To avoid phase delay caused by filtering, we use the forward–backward filtering method that does not introduce phase distortion [33]. Then, the signal within the time interval $[n_i - N, n_4 + N]$ is processed using the HT to obtain the IF. Finally, the relative displacement is calculated using the IF estimates from the interval $[n_i, n_4]$.

From the above analysis, it can be seen that the IFI method is minimally affected by noise intensity and does not accumulate significant errors even over longer estimation durations. Meanwhile, similar to the Vernier method, the IFI algorithm can achieve a resolution beyond the wavelength limit within a single sampling interval c/f_s. Furthermore, when the frequency noise follows a well-behaved zero-mean distribution, the IFI algorithm can attain extremely high accuracy. These characteristics indicate that the IFI algorithm is well-suited for RDE in mobile target localization scenarios where the signal-to-noise ratio (SNR) is affected by position and the estimation time interval is unknown.

C. DSC-MC for Precise ToA Estimation

In this section, we introduce the DSC-MC algorithm, which is based on the frequency-modulated continuous wave (FMCW) method [10], [24] and enables unbiased ToA estimation.

First, we start with a single chirp signal to analyze the error in the FMCW method caused by target motion. Assuming the chirp signal transmitted by the anchor is

$$u_t(t) = \cos 2\pi \left(f_c + \frac{B}{2T}t\right) t \tag{21}$$

where f_c is the starting frequency of the chirp, B is the signal bandwidth, and T is the duration of the signal. Assuming the signal of (21) propagates over a distance R before reaching the target, the received signal is

$$u_r(t) = \alpha \cos 2\pi \left(f_c + \frac{B}{2T}(t - t_d)\right)(t - t_d). \tag{22}$$

The α is the gain of $u_r(t)$, t_d is the delay time. The FMCW method obtains the frequency difference signal $u_\Delta(t)$ by multiplying the transmitted signal $u(t)$ with the received signal $u_r(t)$ and applying a low-pass filter, expressed as

$$u_\Delta(t) = \frac{\alpha}{2} \cos\left(2\pi f_c t_d + \pi \frac{B}{T}\left(2tt_d - t_d^2\right)\right). \tag{23}$$

As stated in [24], when the sound source is stationary and the target moves at a velocity v relative to the sound source, we have $R + vt = ct_d$. And the frequency difference signal becomes

$$\frac{\alpha}{2} \cos\left(2\pi f_c \frac{R + vt}{c} + \frac{2\pi Bt(Rt + v)}{cT} - \frac{\pi B(R + vt)^2}{c^2 T}\right). \tag{24}$$

Taking the derivative of the phase of this signal to obtain the frequency and ignoring $1/c^2$ terms, we get

$$f_\Delta = \frac{1}{2\pi}\frac{\delta \text{Phase}}{\delta t} = \frac{BR}{cT} + \frac{f_c v}{c} + \frac{Bv}{c}. \tag{25}$$

In early researches [13], [14], [15], a downchirp with the same bandwidth B and duration T was introduced to compensate for the velocity-induced deviation, and the frequency difference of the downchirp is

$$f'_\Delta = \frac{BR}{cT} - \frac{f'_c v}{c} + \frac{Bv}{c}. \tag{26}$$

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

The mean value of the frequency differences then is

$$\frac{f_\Delta + f'_\Delta}{2} = \frac{BR}{cT} + \frac{\left(f_c - f'_c\right) v}{2c} + \frac{Bv}{c}. \tag{27}$$

Apparently, by appropriately adjusting f'_c, the deviation term caused by velocity can be reduced. However, such methods have two drawbacks: 1) as shown in Fig. 2, the signal patterns of the traditional methods cannot completely eliminate the velocity term of $(f_\Delta + f'_\Delta/2)$ and 2) the introduced downchirp causes the TFO to double compared with the original signal.

In the following, we explore the achievement of unbiased velocity compensation while maintaining a small TFO.

Our first approach uses two parallel chirp signals to achieve DSC, as shown in Fig. 6(b). When using the same transmitted signal as the reference signal, two frequency differences can be obtained as

$$\begin{aligned} f_\Delta &= \frac{BR}{cT} + \frac{f_c + B}{c} v \\ f'_\Delta &= f_\delta + \frac{BR}{cT} + \frac{f'_c + B}{c} v. \end{aligned} \tag{28}$$

In the equation, f_δ is the frequency difference between parallel chirps. To rewrite this equation, we denote the following matrix and vectors as

$$\mathbf{Q} = \begin{bmatrix} \frac{B}{cT} & \frac{f_c+B}{c} \\ \frac{B}{cT} & \frac{f'_c+B}{c} \end{bmatrix}, \mathbf{f} = \begin{bmatrix} f_\Delta \\ f'_\Delta - f_\delta \end{bmatrix}, \mathbf{x} = \begin{bmatrix} R \\ v \end{bmatrix}. \tag{29}$$

Then, the unknown value $\mathbf{x}$ can be solved by

$$\mathbf{x} = \mathbf{Q}^{-1}\mathbf{f}. \tag{30}$$

As a result, based on the two parallel signals, we can obtain not only an unbiased distance estimate but also the target's velocity relative to the sound source. In addition, parallel chirps have a smaller TFO, with the ratio relative to the original signal given by

$$\text{ratio}_{\text{TFO}} = 1 + \frac{f_\delta}{B}. \tag{31}$$

By adjusting f_δ between the chirps, we can achieve a more flexible TFO compared with the traditional methods.

This approach can also incorporate additional parallel chirps, trading off TFO for improved interference resistance and higher accuracy, as shown in Fig. 6(c). In this case, with N chirps, the parameters are

$$\mathbf{Q} = \begin{bmatrix} \frac{B}{cT} & \frac{f_c+B}{c} \\ \vdots & \vdots \\ \frac{B}{cT} & \frac{f_c^{N-1}+B}{c} \end{bmatrix}, \mathbf{f} = \begin{bmatrix} f_\Delta \\ \vdots \\ f_\Delta^{N-1} - \sum f_\delta \end{bmatrix}, \mathbf{x} = \begin{bmatrix} R \\ v \end{bmatrix}. \tag{32}$$

Since we still have $\mathbf{Qx} = \mathbf{f}$, we can use the least-squares (LS) method to solve for $\mathbf{x}$

$$\begin{aligned} \epsilon &= (\mathbf{Qx} - \mathbf{f})^{\mathrm{T}} (\mathbf{Qx} - \mathbf{f}) \\ \mathbf{x} &= \arg\min_{\mathbf{x}} \epsilon. \end{aligned} \tag{33}$$

And the TFO ratio is

$$\text{Ratio}_{\text{TFO}} = 1 + \frac{\sum f_\delta}{B}. \tag{34}$$

The value of f_δ not only affects the signal's TFO but also impacts estimation accuracy. The influence of f_δ and chirp number n will be analyzed in detail in the subsequent simulations.

The TFO ratio of parallel chirps is still greater than 1. To further reduce this value, the above method can be extended to two different chirps of any form. Since all the signal parameters are known, the coefficient matrix can be expressed as

$$\mathbf{Q} = \begin{bmatrix} \frac{B_1}{cT_1} & \frac{f_{c_1}+B_1}{c} \\ \frac{B_2}{cT_2} & \frac{f_{c_2}+B_2}{c} \end{bmatrix}, \mathbf{f} = \begin{bmatrix} f_{\Delta_1} \\ f_{\Delta_2} \end{bmatrix}, \mathbf{x} = \begin{bmatrix} R \\ v \end{bmatrix}. \tag{35}$$

Both $\mathbf{x} = \mathbf{Q}^{-1}\mathbf{f}$ and LS method can be used to solve for $\mathbf{x}$. Based on this method, we have designed three different signal schemes, each with $\text{ratio}_{\text{TFO}} = 1$. It is worth noting that when the chirp frequency decreases over time (downchirp), the value of f_c becomes negative.

The proposed signal schemes are illustrated in Fig. 6(d)–(f). In Fig. 6(d), the additional signal has the same bandwidth but a shorter duration. This design ensures a TFO ratio of 1. However, the reduced duration not only compromises the accuracy of a single TDE but also increases the risk of unequal velocities between the two signals. In contrast, the signal in Fig. 6(e) shortens only the bandwidth, sacrificing the TDE accuracy of the additional signal. In Fig. 6(f), the additional signal is a downchirp. Unlike the two upchirps, a downchirp can retain the full duration and bandwidth, thereby maximizing the estimation accuracy.

Theoretically, all the schemes using DSC-MC can achieve unbiased estimation. To provide guidance for practical applications, the impact of different parameters in each signal scheme on estimation accuracy will be further explored in subsequent simulations.

IV. Simulation

A. IFI Performance Simulation

In this section, the relationship between IFI performance and SNR as well as length of interval will be quantitatively simulated. These results will serve as a reference for setting various experimental parameters.

The IFI method is also compared with the STFT-based method, the LLAP method [25], and the Vernier method [27]. In the following simulations, a sinusoidal signal with a frequency of 16 kHz is transmitted toward the target, with a sampling rate of 48 kHz and the speed of sound set to 343 m/s. It is assumed that the target is moving away from the sound source. To simulate the randomness of target speed variations during actual measurements, a series of random velocity values are generated within the range of 0–4 m/s. Considering the continuity of motion, these random values are made to increase continuously over time, i.e., we have

$$v[n] \in [0, 4], v[n] \le v[n+1], n = 1, 2, \ldots, N \tag{36}$$

where N is the length of window. Thus, the received signal within the observation window is

$$u_s(n) = \cos 2\pi f \frac{\sum_{i=1}^{n} v[i] \cdot 1/f_s}{c}. \tag{37}$$

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

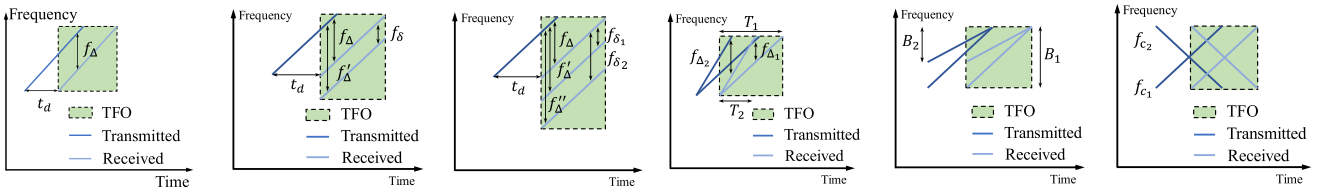


Fig. 6. FMCW method and the five signal schemes proposed based on DSC-MC. (a) FMCW. (b) Scheme A. (c) Scheme B. (d) Scheme C. (e) Scheme D. (f) Scheme E.

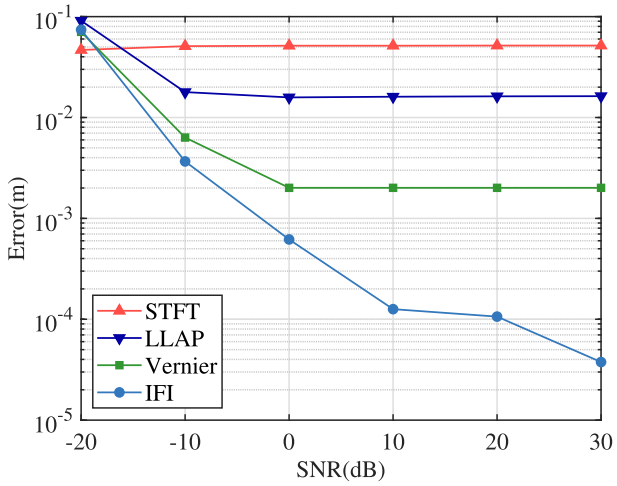


Fig. 7. Simulation results of RDE error under different SNRs, with an estimation duration of 100 ms.

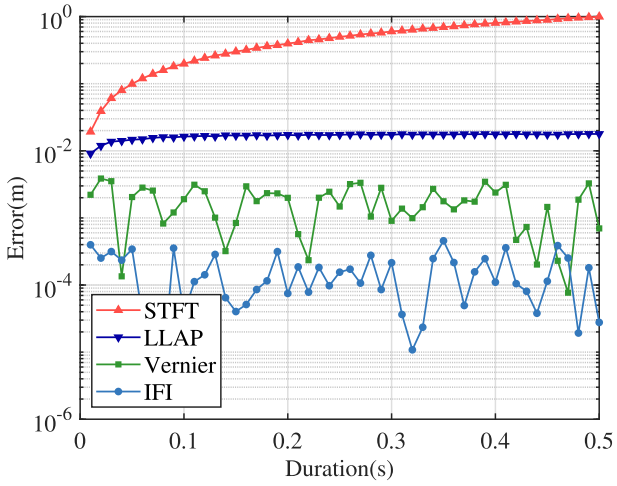


Fig. 8. Simulation results of RDE error under different durations, with an SNR of 10 dB.

In the simulations of this section, each set of simulations with the same parameters is repeated $M = 100$ times, with each run independently generating a set of velocities and the corresponding received signals. The error is calculated as the mean absolute value and is expressed as

$$\text{Error} = \frac{1}{M}\sum_{m=1}^{M}\left|\hat{R}_m - \sum_{n=1}^{N}\frac{v[n]}{f_s}\right|. \tag{38}$$

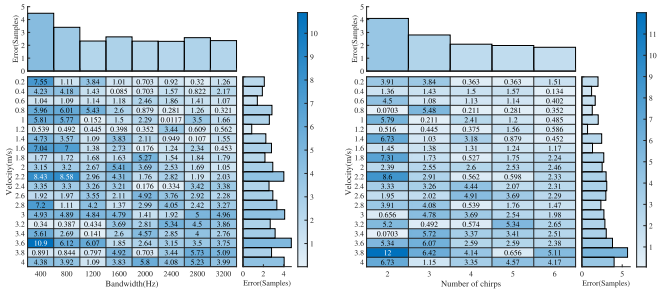
We first test the accuracy of each method under different SNRs. To control the SNR, mutually independent white Gaussian noise was properly scaled and added to each signal. In this simulation, the length of the observation window is set to 4800 samples, corresponding to a duration of 100 ms. The simulation results are shown in Fig. 7. The accuracy of the STFT method remains low across all the SNR levels, with a minimum error of 46.72 mm. In contrast, the accuracy of the other methods improves with increasing SNR. At 0 dB, the LLAP method achieves an accuracy of 15.82 mm, and the Vernier method reaches 2.01 mm. Beyond this point, even as the SNR increases, their accuracy no longer improves significantly. However, at 0 dB, the proposed IFI method already achieves an accuracy of 0.62 mm. When the SNR reaches 30 dB, the accuracy of the IFI method improves further to 3.77×10^{-2} mm. This simulation result is consistent with theoretical expectations, indicating that the IFI method has superior noise robustness. In addition, since the influence of noise on IF has zero mean statistically, the IFI method offers the highest resolution.

In AIPSs, the time interval between ARSs is not fixed. Considering that AIPSs typically have an update rate equal to or greater than 1 Hz, this interval may vary between 0 and 500 ms. With a simulation step size of 10 ms and a maximum observation window length of 500 ms, the simulation results are shown in Fig. 8. In this simulation, to isolate the effect of time interval variations, all the signals are set with an SNR of 10 dB. As shown in the figure, the errors of the STFT-based method and the LLAP method gradually increase with the lengthening of the time interval. In contrast, the errors of the Vernier and IFI methods show no significant correlation with the interval length. The IFI method yields an error of 0.40 mm at a 10-ms interval and 2.77×10^{-2} mm at a 500-ms interval. Across all the interval lengths, the maximum error of the IFI method is only 0.46 mm. This result demonstrates that the IFI method is capable of meeting displacement estimation requirements over varying time intervals.

B. DSC-MC Performance Simulation

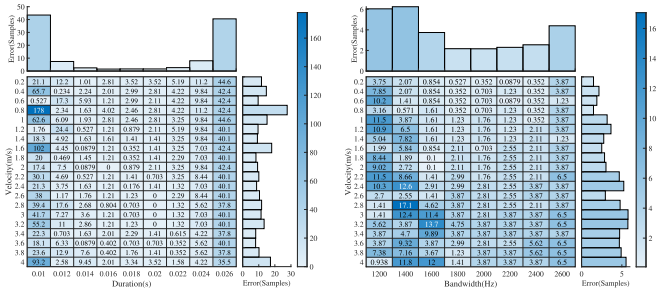
In this section, we first analyze the performance of the five proposed signal schemes under different parameter settings through simulations. Then, under the fixed parameters, all the signal schemes will be compared with traditional methods under varying SNR and velocity conditions. These simulation results will provide guidance for the selection of schemes and parameters in practical applications. For consistency, we continue to use the alphabetical labels introduced earlier for each signal scheme. We define one of the signals in all the signal

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 9. DSC errors of signal Schemes A and B using the DSC-MC method with varying velocities. (a) Scheme A with different f_δ. (b) Scheme B with different N.



(a) (b)

Fig. 10. DSC errors of signal Schemes C and D using the DSC-MC method with varying velocities. (a) Scheme C with different T_2. (b) Scheme D with different B_2.

schemes as an upchirp signal with a starting frequency of 16 kHz, an ending frequency of 19 kHz, and a duration of 30 ms. The parameters of the other signal are adjusted based on this reference signal. All the signals are set at an SNR level of 30 dB.

For Scheme A, which uses two parallel chirp signals, the frequency difference between the two chirps determines the parameters of the second chirp. In this experiment, we use a step size of 400 Hz, up to a maximum of 3200 Hz. The simulation results are shown in Fig. 9(a), where the bars on both sides represent the corresponding mean values. From the perspective of frequency difference, this scheme achieves a lower error level when the frequency difference exceeds 1200 Hz. In terms of speed, variations in velocity have no significant impact on this scheme. For example, at a frequency difference of 1200 Hz, the error is 3.84 sampling points when the speed is 0.2 m/s, 2.67 sampling points at 2 m/s, and 1.09 sampling points at 4 m/s.

Scheme B focuses on the number of signals as the primary factor. In this simulation, the frequency difference between each parallel chirp is fixed at 600 Hz. The simulation results are shown in Fig. 9(b). From the perspective of the number, the average error decreases as the number of parallel chirps increases. When there are two chirps, the error is 4.09 sampling points; when there are six chirps, the error drops to only 1.85 samples. From the perspective of speed, the error increases as speed rises, and this trend is not eliminated by increasing the number of chirps. These results indicate that using multiple chirps based on the LS method can improve accuracy, albeit at the cost of a raised TFO ratio.

In Scheme C, the variable is the duration of the second chirp. Fig. 10(a) shows that as the duration gradually increases, the average error first decreases and then increases. This is because a duration that is too short leads to higher estimation errors for the chirp itself, while a longer duration makes the two signals too similar, causing even small estimation errors to deviate the solution from the true value. When the duration is 16, 18, and 20 ms, the average error of Scheme C is only 1.49, 1.59, and 1.58 samples, respectively. These results can serve as a reference for signal design. In addition, this scheme does not show a clear correlation with velocity.

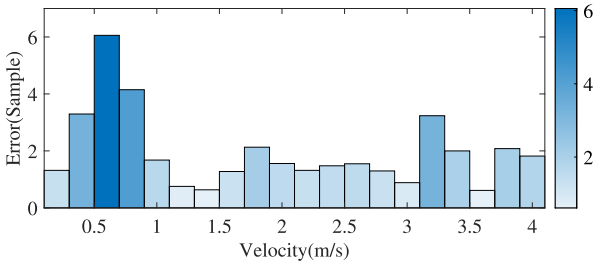


Fig. 11. DSC errors of signal Scheme E using the DSC-MC method with varying velocities.

In Scheme D, the two signals are distinguished by reducing the bandwidth of one signal. As illustrated in Fig. 10(b), as the bandwidth increases, similar to Scheme C, the TDE error first decreases and then increases. The average error reaches its minimum, 2.16 samples, when the bandwidth is 2000 Hz. Scheme D exhibits larger average errors at higher speeds, with an average error of 5.52 samples when the speed is 4 m/s.

Scheme E uses a fixed downchirp signal, so only the effect of speed is considered, as shown in Fig. 11. At a speed of 0.6 m/s, Scheme E exhibits the highest error of 6.06 samples, while at 3.6 m/s, the error reaches its minimum at 0.61 samples. Overall, the scheme shows a relatively small average error of just 1.93 samples.

Based on the simulations presented above, we selected appropriate parameters for the proposed signal schemes and compared them with traditional methods under varying SNRs and velocities. We will compare the proposed method with the raw general cross correlation (GCC), DSC-FMCW [13], and DSC-GCC [15]. The selected parameters for all the signals are listed in Table I. For fairness, all the schemes include chirp signals with similar bandwidths and durations. The simulation results are shown in Fig. 12. For better readability of scale, pure white squares or missing bars in the figure indicate errors that are too large to display ($>$ 80 sampling points).

From the perspective of SNR, it can be observed that most methods fail at 0 dB. Taking a 48-kHz sampling rate as an example, an error exceeding 80 sampling points corresponds to a distance estimation error of more than 0.57 m. This is because all the methods rely on the raw estimation from FMCW or GCC.

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

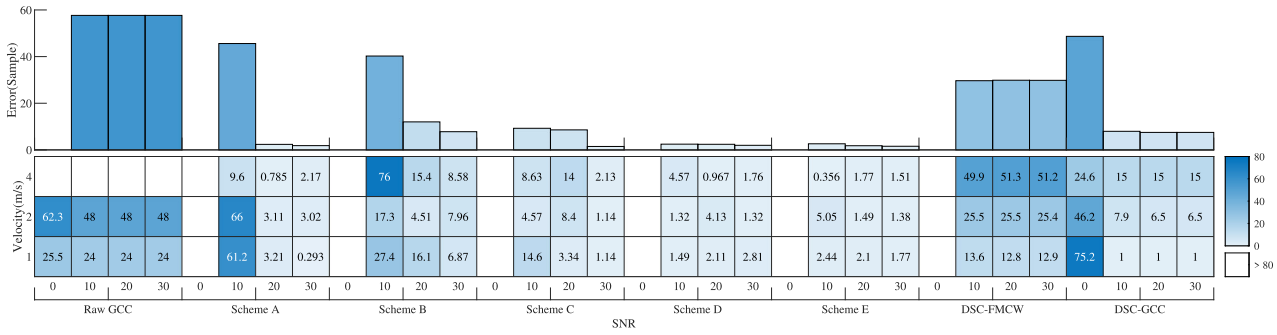


Fig. 12. ToA estimation errors of different methods and signal schemes under various SNRs and velocities.

TABLE I
PARAMETER SETTING

Method	Parameters			$\text{Ratio}_{\text{TFO}}$
	f_c(kHz)	B(kHz)	T(ms)	
Raw GCC	16	3	30	1
Scheme A	(16,17.2)	(3,3)	(30,30)	1.4
Scheme B	(16,17,18)	(3,3,3)	(30,30,30)	1.67
Scheme C	(16,16)	(3,3)	(30,18)	1
Scheme D	(16,17.2)	(3,1,8)	(30,30)	1
Scheme E	(16,19)	(3,-3)	(30,30)	1
DSC-FMCW	(16,19)	(3,-3)	(30,30)	2
DSC-GCC	(19,19)	(3,-3)	(30,30)	2

From the perspective of velocity, the estimation errors of traditional methods increase with speed. Taking DSC-GCC as an example, under SNRs ranging from 10 to 30 dB, all the errors remain one sample at 1 m/s but rise to 15 samples at 4 m/s. This is primarily due to the incomplete cancellation of the velocity term in the estimation formulas used by these methods. However, the proposed methods show no such trend, suggesting that the estimation is effectively decoupled from velocity.

Among the proposed methods, Schemes C, D, and E exhibit not only lower TFO ratios but also superior performance under low SNR conditions. Scheme A has lower errors under SNRs of 20 and 30 dB, with average errors of 2.37 and 1.83 samples, respectively. Under SNRs above 10 dB, Schemes C, D, and E show maximum errors of 9.26, 2.40, and 2.62 samples, and minimum errors of 1.47, 1.96, and 1.56 samples, respectively.

The simulation results indicate that the DSC-MC method offers flexibility in signal design, thereby reducing the TFO. Moreover, under varying speeds and SNRs, certain signal schemes demonstrate superior performance.

V. EXPERIMENT

In this section, extensive experiments are conducted to evaluate the accuracy of the proposed algorithms in real-world scenarios and their performance compared with traditional methods. In all the following experiments, JBL GO2 Bluetooth speakers with a rated power of 3.1 W are used as sound sources or localization anchors, and one digital MEMS silicon microphone is used to receive acoustic signals. The signal

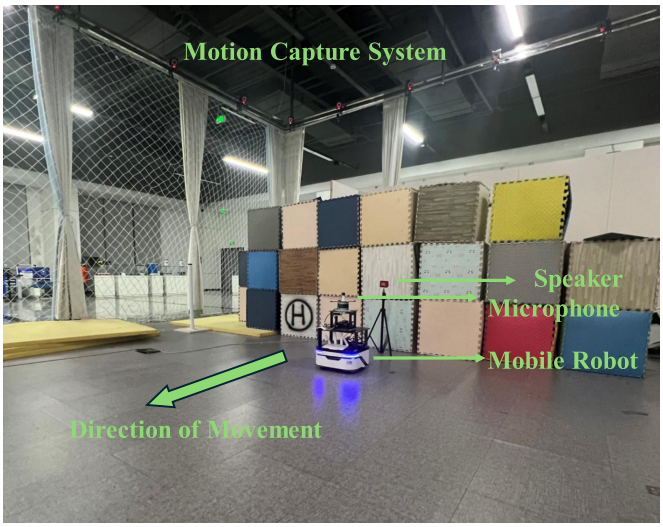


Fig. 13. Experimental setup and environment for testing IFI and DSC-MC algorithms.

parameters in each experiment follow the design specified in Table I. The NOKOV motion capture system is used to provide ground truth for the moving object. The speed of sound is set at 343 m/s.

A. IFI Performance

In this section, the performance of the IFI algorithm is analyzed from both distance and time interval perspectives. The experimental setup and environment are shown in Fig. 13. A microphone is mounted on a mobile robot, which moves away from the speaker along the direction indicated by the arrow. The speaker continuously emits a 16 kHz single-tone sinusoidal signal. The overhead motion capture system recorded the trajectory of the microphone, providing ground truth for RDEs over different distances and intervals.

Fig. 14 shows the mean absolute error (MAE) of RDE at different distances from the sound source, with time intervals ranging from 0.1 to 0.5 s (in 0.1 s steps). As shown in the figure, the proposed IFI algorithm consistently achieves the best performance across all the distances. At a distance of 8 m, the MAE is 0.0033 m; at 10 m, the error is only 0.0042 m. Although these results are slightly worse than the simulation outcomes, they still represent a state-of-the-art performance level. The primary reasons for this discrepancy

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

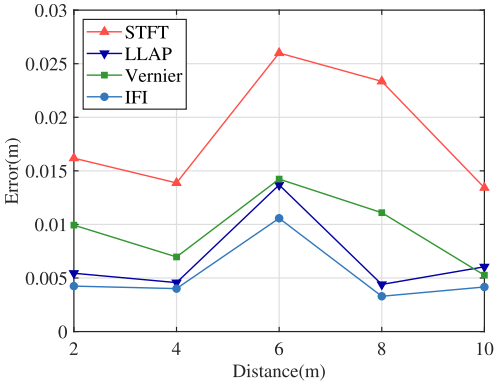


Fig. 14. Average RDE error at different distances.

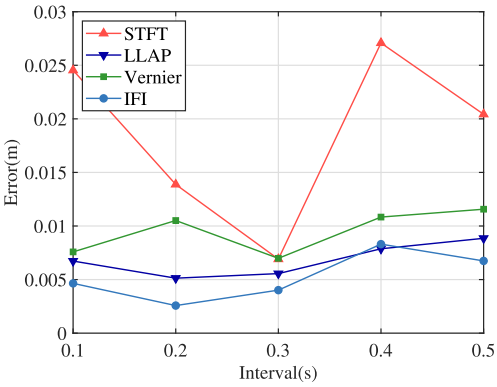


Fig. 15. Average RDE error under different durations.

are twofold: 1) nonideal noise distribution and 2) measurement constraints and hardware-induced errors. The high precision of the IFI algorithm rests on the assumption of zero-mean noise distribution. This condition is difficult to achieve experimentally, where noise characteristics inevitably deviate from theoretical and simulation designs. Furthermore, systematic biases arising from sound speed, microphones, and speakers can hardly be eliminated in experiments. The accuracy of IFI also approaches the resolution limit of the ground-truth system.

Fig. 15 presents the MAE results averaged across all the distances for each time interval. The IFI algorithm exhibits the highest MAE of 0.0083 m at the 0.4-s interval, and the lowest MAE of only 0.0026 m at the 0.2-s interval. These results demonstrate that the IFI algorithm achieves optimal performance in RDE, with errors consistently remaining at the millimeter level. Moreover, the IFI algorithm shows strong robustness across different distances and time intervals, making it well-suited to meet the enhancement needs of long-range, time-uncertain AIPSs.

B. DSC-MC Performance

In this section, the performance of the DSC-MC algorithm will be evaluated. We use Scheme E, which demonstrated the best performance in the simulation, as the signal configuration for DSC-MC in experiments. The experimental setup remains as shown in Fig. 13. In three separate trials, we allow the mobile robot to move at different speeds. Starting from a stationary position, the speaker emits chirp signals at fixed intervals.

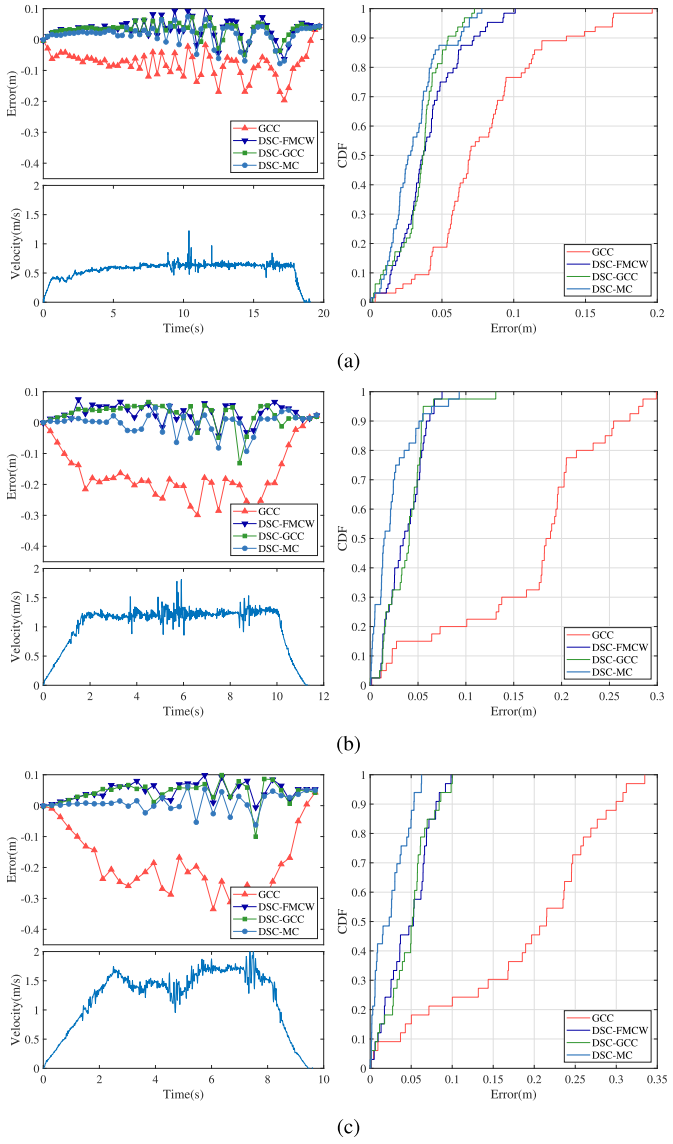


Fig. 16. Curve of distance estimation error versus target velocity, and the corresponding overall cdf of the error. Mean velocity: (a) 0.63, (b) 1.20, and (c) 1.57 m/s.

The results of the three trials are shown in Fig. 16. In each subplot, the upper and lower parts on the left side represent the curves of estimation error and target speed over time, respectively, while the right side shows the cdf of the MAE statistics. Two clear conclusions can be drawn from this figure. First, the traditional GCC algorithm, which does not perform Doppler compensation, exhibits an increasing error with increasing speed. This trend is especially evident in Fig. 16(c), where the average speed is the highest. Second, all the Doppler compensation algorithms effectively reduce the ToA estimation error. As observed from the cdf curves in all the subplots, the DSC-MC algorithm achieves the best performance with an even smaller TFO.

The detailed statistical results are shown in Table II, where the best results under each metric for every experiment are highlighted in bold. It can be seen that the DSC-MC algorithm demonstrates superior performance across different speeds and evaluation metrics. When the average speed reaches 1.57 m/s,

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
STATISTICAL RESULTS OF DISTANCE ESTIMATION ERRORS BY DIFFERENT METHODS

Mean Speed	Method	MAE	Median 50%	90%
0.63 (m/s)	GCC	0.0792	0.0695	0.1196
	DSC-FMCW	0.0406	0.0367	0.0711
	DSC-GCC	0.0354	0.0370	**0.0536**
	DSC-MC	**0.0308**	**0.0273**	0.0613
1.20 (m/s)	GCC	0.1669	0.1861	0.2543
	DSC-FMCW	0.0363	0.0353	0.0587
	DSC-GCC	0.0380	0.0408	0.0534
	DSC-MC	**0.0228**	**0.0150**	**0.0510**
1.57 (m/s)	GCC	0.1875	0.2150	0.2876
	DSC-FMCW	0.0468	0.0527	0.0798
	DSC-GCC	0.0478	0.0515	0.0787
	DSC-MC	**0.0237**	**0.0233**	**0.0503**

* The unit of numbers in the last three columns is meters.



Fig. 17. Experimental setup and environment for positioning tests.

the DSC-MC algorithm still maintains an MAE of 0.0237 m, a median error of 0.0233 m, and a 90th percentile error of 0.0503 m. These experiments and results indicate that the DSC-MC algorithm achieves more effective Doppler compensation with a lower TFO and exhibits excellent robustness across different motion speeds.

C. PALM Performance

In this section, we evaluate the performance of the PALM system through experiments. For fairness, all the localization algorithms adopt the TDoA-ML method for position estimation. The experimental setup and environment are illustrated in Fig. 17. Trajectory tracking is conducted within an area of 6.20 × 5.91 m^2, with four anchors placed at the vertices of a rectangular region. Due to mutual interference among the signal schemes used by different algorithms, each algorithm was tested in a separate trajectory experiment, except for the GCC algorithm, which reused the DSC-MC signal. The three sets of experimental trajectories and localization results are shown in Fig. 18. The localization update rate for all the methods was 2 Hz. During the experiments, the target moved at a randomly varying speed, with the overall

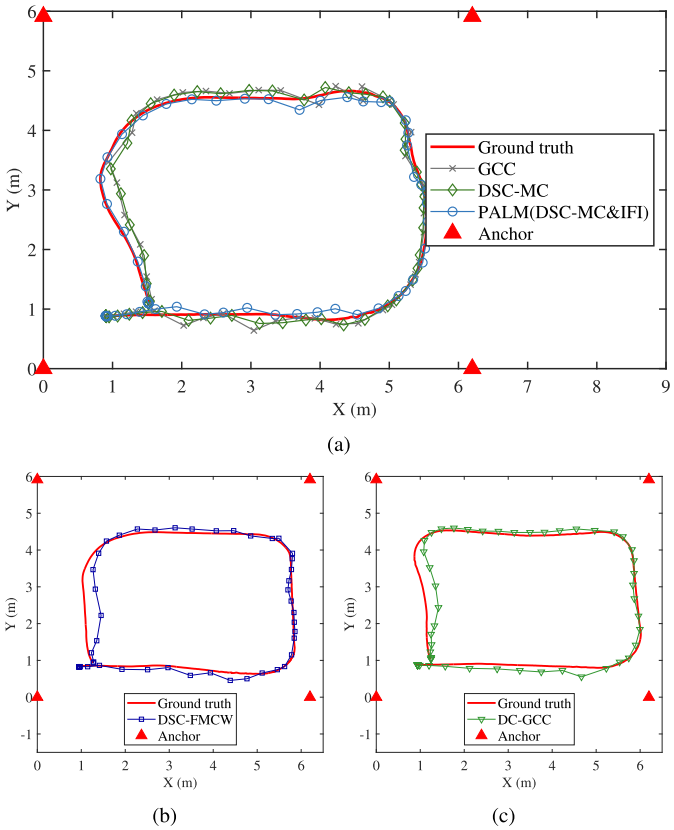


Fig. 18. Trajectory estimations of different methods and the corresponding ground truth. (a) Trajectory 1. (b) Trajectory 2. (c) Trajectory 3.

TABLE III
STATISTICAL RESULTS OF POSITIONING ERRORS BY DIFFERENT METHODS

Method \ Metric	Mean	Median 50%	90%
GCC	0.1562	0.1686	0.3400
DSC-FMCW	0.0782	0.0476	0.1750
DSC-GCC	0.0704	0.0533	0.1530
DSC-MC	0.0915	0.0821	0.1884
PALM (DSC-MC & IFI)	**0.0582**	**0.0335**	**0.1408**

* The unit of all numbers is meters.

trajectory and speed variation kept similar across the three experiments.

From Fig. 18, it is clearly observed that when the localization points are sparse (the target is moving at high speed), algorithms that only perform Doppler compensation exhibit significant deviations. More specifically, Fig. 19 shows the curves of velocity and localization error over time for each algorithm. In Fig. 19, whenever the target speed reaches approximately 1.5 m/s, all the algorithms except PALM experience large localization errors. The maximum localization error for GCC reaches 0.550 m, while DSC-MC, DSC-FMCW, and DSC-GCC have maximum errors of 0.377, 0.364, and 0.299 m, respectively. This indicates that the relative displacement errors discussed previously were not eliminated, so even though Doppler effects were compensated for, localization accuracy remains difficult to improve.

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

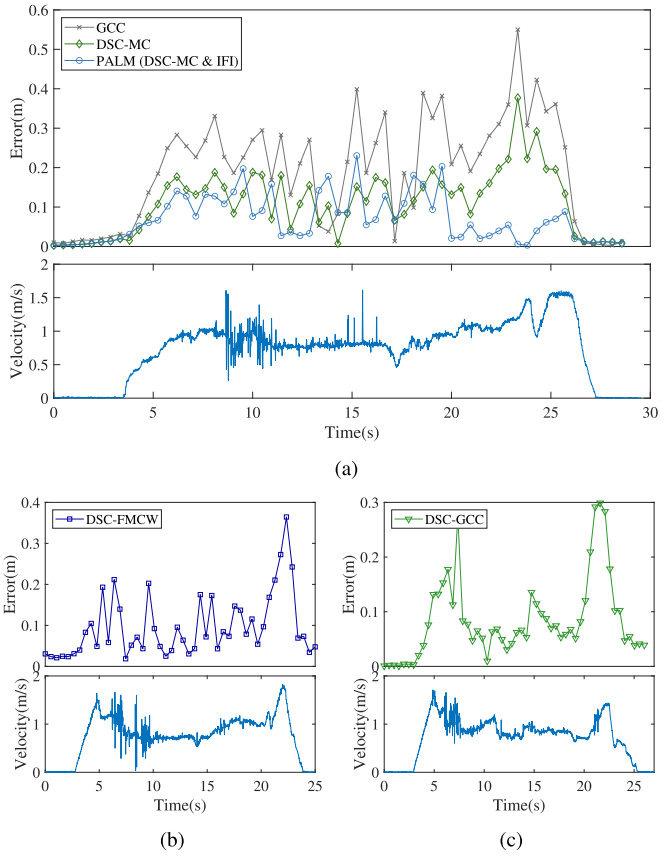


Fig. 19. Curves of positioning error versus target velocity for different methods. Mean velocity: (a) 0.70, (b) 0.71, and (c) 0.70 m/s.

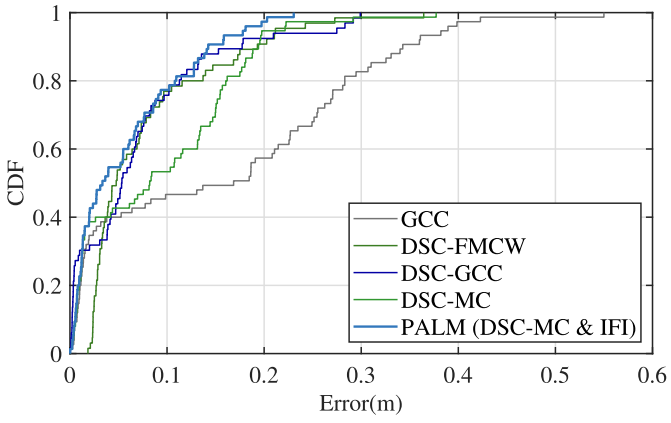


Fig. 20. CDF curves of the positioning error.

Table III records the statistical results of the localization errors, which show that localization algorithms with Doppler compensation outperform the basic GCC method. And Fig. 20 is the CDF plot of the errors of each algorithm, which shows the full picture of the error distribution. The PALM method, which compensates for RDE-related errors, achieves the best accuracy, with a mean localization error of 0.0582 m, a median error of only 0.0335 m, and a 90th percentile error of 0.1408 m. And its mean localization accuracy is improved by 62.7% compared with the basic GCC method, and by 25.6%, 17.3%, and 36.4%, respectively, compared with the other Doppler compensation algorithms.

VI. Conclusion

This article proposes the PALM system, enabling the AIPS to have better positioning performance when facing moving targets. To address two major challenges in mobile target localization, Doppler-shift and the displacement during SRI, the proposed PALM system uses two novel methods: DSC-MC and IFI. Both the simulation and experimental results demonstrate that the DSC-MC method achieves superior estimation accuracy under lower TFO, while the IFI algorithm delivers state-of-the-art performance in RDE. Finally, experiments show that PALM achieves an average localization accuracy of 0.0582 m for targets moving at an average speed of 0.7 m/s, effectively meeting the accuracy requirements of mobile target localization.

References

[1] S. Wang, "Research on indoor UAV swarm positioning technology based on UWB," in *Proc. 8th Int. Conf. Inf. Syst. Eng. (ICISE)*, Jun. 2023, pp. 477–480.

[2] J. Choi, G. Lee, S. Choi, and S. Bahk, "Smartphone based indoor path estimation and localization without human intervention," *IEEE Trans. Mobile Comput.*, vol. 21, no. 2, pp. 681–695, Feb. 2022.

[3] Y. Lin, K. Yu, F. Zhu, J. Bu, and X. Dua, "The state of the art of deep learning-based Wi-Fi indoor positioning: A review," *IEEE Sensors J.*, vol. 24, no. 17, pp. 27076–27098, Sep. 2024.

[4] C. Cai, R. Zheng, and J. Luo, "Ubiquitous acoustic sensing on commodity IoT devices: A survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 432–454, 1st Quart., 2022.

[5] J. Singh, N. Tyagi, S. Singh, F. Ali, and D. Kwak, "A systematic review of contemporary indoor positioning systems: Taxonomy, techniques, and algorithms," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34717–34733, Nov. 2024.

[6] J. Aparicio, F. J. Álvarez, Á. Hernández, and S. Holm, "A survey on acoustic positioning systems for location-based services," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–36, 2022.

[7] Z. Liu, T. Hakala, J. Hyyppä, A. Kukko, and R. Chen, "Performance comparison of UWB IEEE 802.15.4z and IEEE 802.15.4 in ranging, energy efficiency, and positioning," *IEEE Sensors J.*, vol. 24, no. 8, pp. 12481–12489, Apr. 2024.

[8] P. S. Farahsari, A. Farahzadi, J. Rezazadeh, and A. Bagheri, "A survey on indoor positioning systems for IoT-based applications," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7680–7699, May 2022.

[9] Z. Liu, R. Chen, F. Ye, G. Guo, Z. Li, and L. Qian, "Improved TOA estimation method for acoustic ranging in a reverberant environment," *IEEE Sensors J.*, vol. 22, no. 6, pp. 4844–4852, Mar. 2022.

[10] N. Jia et al., "Robust acoustic TOA estimation based on multipath extraction in frequency domain," in *Proc. 13th Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Sep. 2023, pp. 1–7.

[11] Z. Li et al., "Dual-step acoustic chirp signals detection using pervasive smartphones in multipath and NLOS indoor environments," *IEEE Internet Things J.*, vol. 11, no. 4, pp. 6494–6507, Feb. 2024.

[12] N. Jia, W. Cui, X. Wang, L. Yang, C. Xue, and Z. Wang, "FDME-RATE: Frequency-domain-based multipath extraction for robust acoustic TOA estimation," *IEEE Internet Things J.*, vol. 11, no. 20, pp. 32656–32669, Oct. 2024.

[13] Y. Zhuang, Y. Wang, Y. Yan, X. Xu, and Y. Shi, "ReflecTrack: Enabling 3D acoustic position tracking using commodity dual-microphone smartphones," in *Proc. 34th Annu. ACM Symp. User Interface Softw. Technol.*, Oct. 2021, pp. 1050–1062.

[14] Z. Liu et al., "Precise, low-cost, and large-scale indoor positioning system based on audio dual-chirp signals," *IEEE Trans. Veh. Technol.*, vol. 72, no. 1, pp. 1159–1168, Jan. 2023.

[15] Z. Liu et al., "Submeter-level ToF-based acoustic positioning of moving objects with chirp-based Doppler shift compensation," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21916–21929, Jun. 2024.

[16] W. Cui et al., "Doppler shift-compensated FMCW algorithm for centimeter-level acoustic single-anchor positioning of moving targets in narrow spaces," *IEEE Sensors J.*, vol. 25, no. 11, pp. 20195–20208, Jun. 2025.

[17] R. Chen et al., "Precise indoor positioning based on acoustic ranging in smartphone," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.

[18] C. Peng, G. Shen, and Y. Zhang, "BeepBeep: A high-accuracy acoustic-based system for ranging and localization using COTS devices," *ACM Trans. Embedded Comput. Syst.*, vol. 11, no. 1, pp. 1–29, Mar. 2012.
[19] K. Liu, X. Liu, and X. Li, "Guoguo: Enabling fine-grained smartphone localization via acoustic anchors," *IEEE Trans. Mobile Comput.*, vol. 15, no. 5, pp. 1144–1156, May 2016.
[20] Y.-T. Wang, J. Li, R. Zheng, and D. Zhao, "ARABIS: An asynchronous acoustic indoor positioning system for mobile devices," in *Proc. Int. Conf. Indoor Position. Indoor Navig. (IPIN)*, 2017, pp. 1–8.
[21] S. Ogiso, K. Mizutani, N. Wakatsuki, and T. Ebihara, "Robust indoor localization in a reverberant environment using microphone pairs and asynchronous acoustic beacons," *IEEE Access*, vol. 7, pp. 123116–123127, 2019.
[22] C. Cai, R. Zheng, J. Li, L. Zhu, H. Pu, and M. Hu, "Asynchronous acoustic localization and tracking for mobile targets," *IEEE Internet Things J.*, vol. 7, no. 2, pp. 830–845, Feb. 2020.
[23] H. Wang, C. Xue, Z. Wang, L. Zhang, X. Luo, and X. Wang, "Smartphone-based pedestrian NLOS positioning based on acoustics and IMU parameter estimation," *IEEE Sensors J.*, vol. 22, no. 23, pp. 23095–23108, Dec. 2022.
[24] W. Mao, J. He, and L. Qiu, "CAT: High-precision acoustic motion tracking," in *Proc. 22nd Annu. Int. Conf. Mob. Comput. Netw.*, New York, NY, USA, Oct. 2016, pp. 69–81.
[25] W. Wang, A. X. Liu, and K. Sun, "Device-free gesture tracking using acoustic signals," in *Proc. 22nd Annu. Int. Conf. Mobile Comput. Netw.*, 2016, pp. 82–94.
[26] Y. Zhang, J. Wang, W. Wang, Z. Wang, and Y. Liu, "Vernier: Accurate and fast acoustic motion tracking using mobile devices," in *Proc. IEEE Conf. Comput. Commun.*, Apr. 2018, pp. 1709–1717.
[27] Y. Liu et al., "Vernier: Accurate and fast acoustic motion tracking using mobile devices," *IEEE Trans. Mobile Comput.*, vol. 20, no. 2, pp. 754–764, Feb. 2021.
[28] Y. Ma and Y. Kong, "FRFT based on joint estimation time delay and radial velocity of underwater target," in *Proc. 3rd Int. Congr. Image Signal Process.*, vol. 9, Oct. 2010, pp. 4074–4078.
[29] P. H. Leong, T. D. Abhayapala, and T. A. Lamahewa, "Multiple 3D far-field/near-field moving target localization using wideband echo chirp signals," *IEEE Trans. Signal Process.*, vol. 62, no. 22, pp. 6003–6015, Nov. 2014.
[30] B. Boashash, "Estimating and interpreting the instantaneous frequency of a signal. I. Fundamentals," *Proc. IEEE*, vol. 80, no. 4, pp. 520–538, Apr. 1992.
[31] R. C. Buck, *Advanced Calculus*. Waveland Press, 2003.
[32] B. Boashash, "Estimating and interpreting the instantaneous frequency of a signal. II. Algorithms and applications," *Proc. IEEE*, vol. 80, no. 4, pp. 540–568, Apr. 1992.
[33] F. Gustafsson, "Determining the initial states in forward-backward filtering," *IEEE Trans. Signal Process.*, vol. 44, no. 4, pp. 988–992, Apr. 1996.

Yuzhang Xi received the B.S. degree in automation from Zhejiang University, Hangzhou, China, in 2019, where he is currently pursuing the Ph.D. degree with the Department of Control Science and Engineering.

His current research interests include acoustic signal processing, indoor positioning, and sensor array localization.

Weimeng Cui received the B.S. degree in information science and technology from Northeastern University, Shenyang, China, in 2022. He is currently pursuing the M.S. degree in control engineering with Zhejiang University, Hangzhou, China.

His research interests include indoor positioning, machine learning, and the Internet of Things.

Guangyao Liu received the B.S. degree from Central South University, Changsha, China, in 2022. He is currently pursuing the Ph.D. degree in control engineering with Zhejiang University, Hangzhou, China.

His research interests include robot navigation, machine learning, and acoustic positioning.

Zhi Wang (Member, IEEE) received the B.E. degree in mechanism from Shenyang Jianzhu University, Liaoning, China, in 1991, the M.Sc. degree in mechanical engineering from Southeast University, Nanjing, Jiangsu, China, in 1997, and the Ph.D. degree in automated control from Shenyang Institute of Automation, China Academy of Science, Liaoning, in 2000.

He has co-authored more than 100 publications in international journals, such as IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, IEEE TRANSACTIONS ON MOBILE COMPUTING, and IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS and conferences. His current research interests include indoor positioning, satellite navigation, and location-based services.

Authorized licensed use limited to: Zhejiang University. Downloaded on November 01,2025 at 04:15:45 UTC from IEEE Xplore. Restrictions apply.